

MCA01 – Due: Fri., Oct. 02, 2026, before 11:50 PM

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%inel4301s999gp99sn99mca02
%Prof. Domingo Antonio Rodriguez
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%*****Prof. D. Rodriguez*****
%*****DSB-SC COMMUNICATIONS SYSTEMS AND GAUSSIAN NOISE*****
%*****ANALOG AND DIGITAL COMMUNICATIONS SIMULATIONS*****
%*****
clear all                                %Clear working space
close all
%*****
disp('*****')
disp('Begin Simulation for Underwater Communications.')
disp('*****')
%*****
%*****Wanted Signal Input/Output*****
%*****
[sig,Fs]=audioread('dtmfad.wav'); %Get overall sampling frequency
save dsig.txt sig -ascii           %Save signal as dsig.txt file
load -ascii dsig.txt;             %Load ascii file dsig.txt
wsiz=length(dsig);                %Get the length of the "dsig" signal
%*****
%*****Parameter Setting*****
%*****
Ts=1/Fs;                           %Sampling time or temporal resolution
Nq=wsiz;                           %Number of samples of wanted signal
Nqseg=round(Nq/10);                %Length of a segment of a signal
Vq=Nq*Ts;                          %Observation window or signal duration
fincq=1/Vq;                        %Frequency resolution of wanted signal
fkq=-(Fs/2):fincq:(Fs/2)-fincq;    %Frequency axis of wanted signal
tq=0:Ts:Vq-Ts;                    %Time axis of wanted signal
%
fV=4000;                           %Cut-off frequency of wanted signal
Wn=(2*fV/Fs);                      %Normalized cut-off frequency
Lp=(1/567);                        %Proportionality factor
Nh=2*floor(Lp*Nq)+1;               %Low-pass filter length
hL=fir1(Nh,Wn);                    %Impulse response of low-pass filter
dsigL=conv(dsig,hL);               %Filtered wanted signal up to fV=4000 Hz
%
f1=4400;                           %Input interference frequency
f2=4900;                           %Input interference frequency
f3=5400;                           %Input interference frequency
g1=cos(2*pi*f1*tq);                %Unwanted input signal
g2=cos(2*pi*f2*tq);                %Unwanted input signal
g3=cos(2*pi*f3*tq);                %Unwanted input signal
g=(1/80)*g1+(1/160)*g2+(1/240)*g3; %Sum of unwanted signals
gmax=max(g);                       %Maximum value of sum
g=transpose((1/gmax)*g);           %Normalized interference signal
Pgseg=1; %Pointer (>0) to the beginning of segment a segment sample
tgseg=(Pgseg-1)*Ts:Ts:(Pgseg+Nqseg-2)*Ts; %Time axis of segment
gseg=g(Pgseg:Pgseg+Nqseg-1); %Selected segment of signal
%*****
%*****SIGNALS AND SYSTEMS MODELING*****
%*****
fc=9900;                           %Carrier frequency
c=transpose((cos(2*pi*fc*tq))); %Carrier signal
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wsig=dsigL(1:Nq,1); %Wanted input signal
Pwsigseg=Nqseg+1; %Pointer (>0) to the beginning of segment a segment sample
twsigseg=(Pwsigseg-1)*Ts:Ts:(Pwsigseg+Nqseg-2)*Ts; %Time axis of segment
wsigseg=wsig(Pwsigseg:Pwsigseg+Nqseg-1); %Selected segment of signal
xm=wsig+g; %Sum of wanted and unwanted signals
xc=xm.*c; %Modulated wanted signal plus interference
%*****
%*****BAND-PASS FILTER*****
Nb=1001; %Order of band-pass filter
Wb=[(2*4640)/Fs (2*12640)/Fs]; %Vector of normilized cut-off freqs.
hB=fir1(Nb-1,Wb,'bandpass'); %Impulse response of band-pass filter
%*****
%*****CHANNEL INPUT SIGNAL*****
yci=conv(xc,hB); %Removal of unwanted interferences
Nyci=length(yci); %Order of channel input signal
tyci=0:Ts:(Nyci-1)*Ts; %Time axis for yci(t)
fryci=1/(Nyci*Ts); %Frequency resolution for Yci(f)
fayci=(-Fs/2):fryci:(+Fs/2)-fryci; %Frequency axis of wanted signal
%*****
%*****CHANNEL NOISE*****
nsig=randn(size(yci)); %Generation of AWGN signal: noise signal
nnsig=(1/max(abs(nsig)))*nsig; %Normalized noise signal
nATT=sqrt(1/24000); %Noise attenuation about -44 dB SNR
nsig=nATT*nnsig; %Attenuated noise present in the channel
%*****
%*****CHANNEL OUTPUT*****
yco=yci+nsig; %Channel output signal with Gaussian noise
tqn=0:(length(yco)-1)*Ts; %New time axis for output signal
cn=transpose((cos(2*pi*fc*tqn))); %New carrier signal
yd=yco.*(2*cn); %Demodulation stage
%*****
%*****LOW-PASS FILTER*****
Nlow=663; %Designed low-pass filter order
Wlow=(2*fV)/Fs; %Normalized cut-off frequency
hlow=fir1(Nlow-1,Wlow); %FIR low-pass impulse response function
taxis_hlow=0:Ts:(length(hlow)-1)*Ts;
fres_hlow=1/((length(hlow))*Ts);
faxis_hlow=(-Fs/2):fres_hlow:(+Fs/2)-fres_hlow;
%*****
%*****RECEIVER OUTPUT*****
xr=conv(yd,hlow); %Low pass filtering at the receiver
taxis_xr=0:Ts:(length(xr)-1)*Ts;
fres_xr=1/((length(xr))*Ts);
faxis_xr=(-Fs/2):fres_xr:(+Fs/2)-fres_xr;
%*****RATE_PARAMETERS*****
%*****
Dr=Fs; %Baud rate is equal to the sampling rate Fs
%*****SPECTRA*****
%*****
fwsig=fft(wsig); %Spectrum of wanted signal (Fund. Reg.)
sfwsig=fftshift(fwsig); %Spectrum of wanted signal (Princ. Reg.)
asfwsig=abs(sfwsig); %Magnitude of spectrum of wanted signal
%
fxc=fft(xc); %Spectrum of modd. signal (Fund. Reg.)
sfxc=fftshift(fxc); %Spectrum of modd. signal (Princ. Reg.)
asfxc=abs(sfxc); %Magnitude of spectrum of modd. signal
%
fyci=fft(yci); %Spectrum of input signal (Fund. Reg.)
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sfcy=fftshift(fyci);           %Spectrum of input signal (Princ. Reg.)
asfcy=abs(sfcy);               %Magnitude of spectrum of input signal
%
fyco=fft(yco);                 %Spectrum of output signal (Fund. Reg.)
sfyco=fftshift(fyco);          %Spectrum of output signal (Princ. Reg.)
asfyco=abs(sfyco);             %Magnitude of spectrum of output signal
%
fyd=fft(yd);                   %Spectrum of output signal (Fund. Reg.)
sfyd=fftshift(fyd);            %Spectrum of output signal (Princ. Reg.)
asfyd=abs(sfyd);               %Magnitude of spectrum of output signal
%
fnsig=fft(nsig);               %Spectrum of noise signal (Fund. Reg.)
sfnsig=fftshift(fnsig);        %Spectrum of noise signal (Princ. Reg.)
asfnsig=abs(sfnsig);           %Magnitude of spectrum of noise signal
%
fhlow=fft(hlow);               %Spectrum of IRF (Fund. Region)
sfhlow=fftshift(fhlow);         %Spectrum of IRF (Princ. Region)
asfhlow=abs(sfhlow);           %Magnitude of spectrum IRF
%
fxr=fft(xr);                   %Spectrum of xr(t) (Fund. Region)
sfxr=fftshift(fxr);            %Spectrum of xr(t) (Princ. Region)
asfxr=abs(sfxr);               %Magnitude of spectrum IRF
%*****PLOTS*****
plot(tq,wsig);
axis([0 (length(wsig)*Ts) min(wsig) max(wsig)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Wanted Signal s(t) in the Time Domain')
grid
%
figure
plot(twsigseg,wsigseg);
%axis([0 (length(twsigseg)*Ts) min(wsigseg) max(wsigseg)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Segment of Wanted Signal s(t) in the Time Domain')
grid
%
figure
plot(tq,g);
axis([0 (length(g)*Ts) min(g) max(g)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Interference Signal g(t) in the Time Domain')
grid
%
figure
plot(tgseg,gseg);
%axis([Pgseg*Ts (Pgseg+length(gseg)*Ts) min(gseg) max(gseg)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Segment of Interference Signal g(t) in the Time Domain')
grid
%
figure
plot(tq,xm);
axis([0 (length(xm)*Ts) min(xm) max(xm)]);
xlabel('Time in Seconds')
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ylabel('Amplitude')
title('Modulating Signal xm(t) in the Time Domain')
grid
%
figure
plot(tq,xc)
axis([0 (length(xc)*Ts) min(xc) max(xc)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Modulated Signal xc(t)')
grid
%
figure
plot(tyci,nnsig)
axis([0 (length(nnsig)*Ts) min(nnsig) max(nnsig)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Normalized Noise Signal nn(t)')
grid
%
figure
plot(tyci,yci)
axis([0 (length(yci)*Ts) min(yci) max(yci)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Channel Input Signal yci(t)')
grid
%
figure
plot(tyci,yco)
axis([0 (length(yco)*Ts) min(yco) max(yco)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Channel Output Signal yco(t)')
grid
%
figure
plot(fkq,asfwsig)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum S(f) of Wanted Signal s(t)')
grid
%
figure
plot(fkq,asfxc)
xlabel('Freq in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Xc(f) of Modulated Signal xc(t)')
grid
%
figure
plot(fayci,asfyici)
xlabel('Freq in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yci(f) of Channel Input Signal yci(t)')
grid
%
figure
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```
plot(fayci,asfyco)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yco(f) of Channel Output Signal yco(t)')
grid
%
figure
plot(fayci,asfyd)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yd(f) of Demodulator Output yd(t)')
grid
%
figure
plot(taxis_hlow,hlow);
axis([0 (length(hlow)*Ts) min(hlow) max(hlow)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Impulse Response Function of Receiver''s Low-Pass Filter')
grid
%
figure
plot(faxis_hlow,asfhlow)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum of Impulse Response Function hlow(t)')
grid
%
figure
plot(taxis_xr,xr);
axis([0 (length(xr)*Ts) min(xr) max(xr)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Receiver''s Low-Pass Filter Output Signal xr(t)')
grid
%
figure
plot(faxis_xr,asfxr)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum of Receiver''s Output xr(t)')
grid
%
%*****OUTPUTS*****
sound(wsig,Fs) %Sound of Wanted Signal
disp('Sound of Wanted Signal')
pause(20)
sound(g,Fs) %Sound of Interference Signal
disp('Sound of Interference Signal')
pause(20)
sound(xm,Fs) %Sound of Modulating Signal
disp('Sound of Modulating Signal')
pause(20)
sound(xc,Fs) %Sound of Modulated Signal
disp('Sound of Modulated Signal')
pause(20)
sound(yci,Fs) %Sound of Channel's Input Signal
disp('Sound of Channel's Input Signal')
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pause(20)
sound(yco,Fs)    %Sound of Channel's Output Signal
disp('Sound of Channel''s Output Signal')
pause(20)
sound(xr,Fs)    %Sound of Channel's Output Signal
disp('Sound of Receiver''s Output Signal')
%*****
disp(['Baud Rate: ', num2str(Dr)])
%*****
disp('*****')
disp('Simulation ended succesfully.')
disp('*****')
```